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METHODS ARTICLE

TRUSTWORTHINESS IN COMPUTATIONAL THEORY CONSTRUCTION: DIMENSIONALIZATION AND CATEGORY SURFACING[1]

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In this methods article, we unpack how researchers can foster trustworthiness in dimensionalization and category surfacing (DCS), a key method family within the genre of computational theory construction (CTC). Information systems (IS), management, and organizational scholars are increasingly leveraging DCS tools such as topic modeling, word embeddings, and clustering to surface latent categories and dimensions from textual data for theory construction. Yet they struggle because evaluations of such research often default to transparency, operationalized as replicability and accountability, which obscures the analytical choices that actually make DCS research rigorous. In this study, we recast transparency as a means toward fostering trustworthiness. We treat researchers' analytical moves as the primary unit of methodological reasoning in how they design, conduct, and disclose their choices across research phases. We develop a framework that authors, reviewers, and editors can use to construct and evaluate DCS research. The framework specifies how trustworthiness arises from the interplay of two research design choices: giving primacy to theoretical versus practice lexicons, and whether the content of texts or the structure of the corpus carries the theoretical load. We articulate expectations for conduct and disclosure across these design choices, clarifying how proportionate reasoning anchors trustworthiness. We conclude with implications for advancing trustworthiness within the broader CTC community and across other computational approaches to research.

Keywords: Computational Theory Construction, Dimensionalization and Category Surfacing, Trustworthiness, Analytical Moves, Computational Tools, Artificial Intelligence

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Introduction

Computational theory construction (CTC) is an emerging research genre that involves the use of machine learning (ML) algorithms and other artificial intelligence (AI) tools to construct theories from large volumes of data (Berente et al., 2019; Miranda et al., 2022a). Within CTC research, computational tools that allow for dimensionalization and category surfacing (DCS)—including topic modeling, word embeddings, and clustering, among others (henceforth *DCS tools*) (see Miranda et al., 2022a)—have become central for interpretive scholars. DCS tools facilitate surfacing latent categories and dimensions from textual corpora, a practice traditionally fundamental to constructing theories through qualitative analysis (Grodal et al., 2021; Strauss & Corbin, 1990). As interest in leveraging DCS tools for constructing theories (henceforth *DCS research*) grows, so do the expectations of transparency to evaluate its rigor. Yet authors, reviewers, and editors face recurring challenges when conducting and evaluating DCS research, many of which stem from treating transparency as an end in itself.

Across scientific fields and leading journals in information systems (IS), management, and organizational research, calls for transparency have amplified (Burton-Jones et al., 2021; DeCelles et al., 2021; see also Aguinis et al., 2024). However, prevailing guidelines are often calibrated to traditional quantitative and qualitative research genres and hence travel poorly to CTC, and to DCS research in particular. Importing quantitative standards presumes stable procedures and externally verifiable replicability, yet unsupervised DCS tools rarely yield a unique model; results are sensitive to preprocessing and parameterization and unsettle any ground truth (Grimmer et al., 2022; Valtonen et al., 2024). Importing qualitative standards presumes researchers can fully account for their decisions, but DCS research embeds algorithmic assumptions and model-internal moves that remain partly opaque to the researchers, complicating accountability (cf., Constantiou et al., 2023; Lebovitz et al., 2021; Waardenburg et al., 2022).

In this methods article, we reframe the conversation by centering trustworthiness and treating transparency as a means to that end. Our methodological review and author interviews show that DCS researchers are often pushed toward conflicting practices and expectations in how to facilitate trustworthiness. We develop a framework that helps researchers design, conduct, and disclose analytical moves for constructing theory with DCS tools in ways that are conceived as trustworthy by reviewers and editors. The framework turns on two research design choices that govern both conduct and disclosure: (1) whether researchers choose to give *primacy to theoretical lexicons* or *primacy to practice lexicons* (i.e., whether the meaning system guiding analysis originates primarily in theory or in the phenomenon), and (2) whether the *content of texts* carry the theoretical load, or the *structure of the corpus* is the focus of theory construction. Positioning a study on these two axes clarifies purpose and appropriate analytical moves against emergent tensions. We then specify expectations across three phases of DCS research, including data work, model building and pattern surfacing, and validating and theorizing, as well as cross-phase iterations. We argue that researchers need to adapt their analytical moves in accordance with their chosen approach to foster trustworthiness.

Our framework offers actionable guidance for authors, reviewers, and editors engaged with DCS research that analytically generalizes to the broader CTC community. For authors, it maps two research design choices—giving primacy to theoretical versus practice lexicons, and a focus on the content of texts versus the structure of the corpus—to phase-specific and iterative analytical moves. For reviewers, it provides a basis to inspect whether those moves are explicit and proportionate to the study's aims. For editors, it supports reviewer assignment and decisions by emphasizing fit among research objectives, analytical moves, and evidence. We also offer implications for a larger CTC and computational research community who may be engaging with different types of computational tools for theory construction or may be using DCS tools to test deductively constructed theories.

From Transparency to Trustworthiness: Expectations of Rigor in DCS Tools and Practices

In this section, we start by explaining how expectations of rigor in IS, management, and organizational research often translate into guidance for achieving transparency. While this guidance spans across qualitative and quantitative genres, with divergent meanings of transparency across these genres, there is little guidance that speaks directly to the CTC research genre and none tailored to the DCS method family. We then explain why treating transparency as an end goal when evaluating rigor fits poorly with DCS research, given the nature of the computational tools and research practices. We close by arguing that instead of attempting to develop specific guidance on transparency for DCS research, the community needs to recast transparency as a means toward fostering trustworthiness in order to integrate concerns for conduct and the disclosure of methods and to improve the fit with exploratory, abductive DCS work.

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Divergent Meanings of Transparency Across Research Genres

Expectations of rigor in research often translate into transparency guidelines (Jacobs et al., 2021; also see Constantinides et al., 2012) with research genre-specific meanings and recommendations. While some discussions on transparency imply a close connection with replicability (e.g., Aguinis & Solarino, 2019; Schooler, 2014), others equate transparency with accountability (e.g., Monteiro et al., 2022; Sarker et al., 2013).

A first set of meanings links transparency to replicability, often interchangeably with reproducibility, especially in the quantitative research genre. Reproducibility concerns whether independent analysts can obtain the same results with the same data and draw similar conclusions; replicability asks whether comparable results can be obtained with new data collected under similar designs and procedures (Aguinis et al., 2018). An underlying assumption that is implicit in this pursuit of replicability is the existence of ground truth, i.e., identification of a data point as a true representation of the world against which the evidence provided by a given study can be evaluated (e.g., Marshall et al., 2023). Though critics have noted limits to replication across contexts and time (e.g., Tremblay et al., 2021).

A second set of meanings equates transparency with accountability, especially in the qualitative research genre. Here, transparency aims to render visible the researcher's positioning and choices so that a knowledgeable reader can audit the chain from data to knowledge claims (Constantinides et al., 2012; Monteiro et al., 2022; Sarker et al., 2013). The underlying assumption in the discourse around accountability is the ability of researchers to systematically account for and, thus, be transparent about their choices and the interpretations that they draw from the data they analyze (Sarker et al., 2013). However, scholars have cautioned against the accountability drive, when some research approaches may be bound by value judgements and may not offer systematic methods (Elliott, 2022). In such cases, "transparency about value judgments in particular could also cause people to develop an unwarranted skepticism about scientific results once they realize that scientific research is not as straightforward as it might initially appear" (Elliott, 2022, p. 349).

DCS Research and the Inadequacy of Transparency as an End

Despite growing guidance on transparency for quantitative and qualitative genres across journals, including *MIS Quarterly* and *Academy of Management Journal*[2] (e.g., Burton-Jones et al., 2021; DeCelles et al., 2021), there is comparatively little guidance for CTC and DCS research. While one could attribute this lack of guidance to the CTC research genre's nascency, we argue for a more fundamental reason: Treating transparency as an end does not travel well into DCS research because of the nature of DCS tools and research practices.

DCS tools represent a class of computational methods and techniques crucial for theory construction in contemporary qualitative and computationally intensive research (see Simeonova & Galliers, 2023). Common DCS tools for text mining and natural language processing include topic modeling, word embeddings, and clustering (see Aceves & Evans, 2023; Hannigan et al., 2019; Kobayashi et al., 2018; Schmiedel et al., 2019). These tools leverage unsupervised algorithms to uncover latent semantic structures within textual data. Rather than starting with predefined categories, these approaches help researchers to inductively identify latent patterns, surfacing novel theoretical dimensions from the data (Miranda et al., 2022a). DCS research proceeds through data work (e.g., sampling and text preprocessing), model building and pattern surfacing (e.g., choosing algorithms as well as parameters), and validation and theorizing (e.g., sensitivity analysis and interpretive validation). These steps involve significant manual labor and qualitative judgment. DCS research is prized in interpretive CTC precisely because it combines manual and automated activity to surface theoretically relevant patterns in text (see the transparency materials[3] for a primer on DCS tools).

The nature of DCS tools makes existing transparency templates ill-posed for DCS research. On the one hand, importing quantitative standards assumes stable procedures and externally verifiable replicability, treating DCS tools as similar to traditional quantitative tools (e.g., fixed effects regression models). This assumption is problematic because unsupervised DCS tools rarely yield a single model that could embody "one right way" of constructing theories (Grimmer et al., 2022, p. 30); outputs are sensitive to choices in preprocessing and parameterization, complicating replicability and the very notion of a ground truth (Grimmer et al., 2022; Valtonen et al., 2024). On the other hand, importing qualitative standards assumes that researchers can fully account for their decisions, treating DCS tools like traditional qualitative analysis tools (e.g., NVivo). But unlike

[2]Also see transparency guidance for other journals: *Management Science* (https://pubsonline.informs.org/page/mnsc/code-and-data-disclosure-policy), *Organization Science* (https://pubsonline.informs.org/page/orsc/data-and-methods-transparency), and *Administrative Science Quarterly* (https://journals.sagepub.com/pb-assets/cmscontent/ASQ/Data%20and%20Methods%20Transparency%20Policy%20040324-1712174442.pdf).

[3] The transparency materials are available at https://osf.io/c9th4/files/xcmzf

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typical qualitative coding exercises for surfacing categories (c.f., Grodal et al., 2021; Strauss & Corbin, 1990), in DCS research with computational tools, many consequential analytical moves occur inside the tools themselves, through automated activity. DCS tools embody algorithmic assumptions and moves that can remain opaque even to expert users, thus complicating accountability (cf., Constantiou et al., 2023; Lebovitz et al., 2021; Waardenburg et al., 2022).

What further complicates generic transparency guidance is that in DCS research, knowledge is created through an iterative process in which researchers are both guided by and actively shape algorithmic outputs. Here, Grodal et al.'s (2021) "active-categorization" perspective is instructive: theory building proceeds through purposeful sequences of moves—asking questions, focusing on puzzles, and dropping, merging, splitting, relating, and sequencing categories for theory development. DCS research naturally instantiates this logic, but with a twist: category structures emerge through a reflexive relationship between the researcher and the algorithm(s) (Günther et al., 2023; Lindberg, 2020). Thus, extant transparency prescriptions are unfit to guide researchers, either because they demand sameness where multiple defensible choices exist, or because they emphasize standardized reporting that obscures how specific moves shaped and were shaped by algorithmic outputs.

Recasting Transparency as a Means To Foster Trustworthiness in DCS Research

To move beyond these impasses, we argue for recasting transparency as a means to foster trustworthiness. We argue that the community may not benefit from trying to develop generic transparency guidelines for DCS research. This is because what seems to be lost in existing debates on replicability and accountability is the important role of transparency in accelerating knowledge production through reflexivity (Günther et al., 2023). In the spirit of cumulative tradition, researchers should reflect on different scenarios and choices and the applicability of different practices to their own research (Harley & Cornelissen, 2022; Pratt et al., 2022).

We thus turn to the construct of trustworthiness (Pratt et al., 2020, 2022). It allows us not to conceive transparency as an end in itself but as something to be calibrated to the study's research objectives and practices, i.e., the specific research choices one makes in terms of which methods to apply and how, and the purpose of employing specific research practices in specific contexts (Monteiro et al., 2022; Pratt et al., 2020, 2022). This orientation is consistent with foundational arguments in qualitative research: Evaluative standards should be fit for purpose and attentive to the distinctive threats and assurances of the research design, rather than imported wholesale from adjacent paradigms (Amis & Silk, 2008; Easterby-Smith et al., 2008; Guba, 1981; Lincoln & Guba, 1985). Put differently, the "degree of transparency here is not set by the demands for replication but is more broadly set by the degree to which authors can convince the reader that they have been honest in how their research has been carried out" (Pratt et al., 2020, p. 12). Trustworthiness also goes beyond accountability because it is not primarily achieved by reporting standardized knowledge production processes or templates but by encouraging researchers to devise "analytical moves tailored to their research objectives" (Pratt et al., 2022, p. 212).

This view aligns with the idea that rigor "does not reside in protocolized techniques or methods but in the very process of researchers reasoning about their use of particular techniques or methods and reporting such reflections in their articles" (Harley & Cornelissen, 2022, p. 240). Accordingly, we treat analytical moves as methodological reasoning, encompassing both methodological conduct (how to design and enact research practices) and disclosure (what to report and how). Previous scholarship on trustworthiness makes this move-centered logic explicit by tying confidence in claims to concrete practices and their documentation—e.g., triangulation, prolonged engagement, peer debriefing, member checks, and audit trails—rather than to compliance with a single reporting template (Guba, 1981; Locke & Ramakrishna Velamuri, 2009). More recent methodological work similarly frames transparency devices (e.g., structured tables) as means of making analytic reasoning inspectable in ways that can strengthen trustworthiness without reducing qualitative work to procedural box-checking (Cloutier & Ravasi, 2021; O'Kane et al., 2021). Thus, fostering trustworthiness does not involve DCS scholars forcing their analyses into inherited transparency templates; rather, it encourages methodological bricolage and analytical, move-centric reasoning aligned with research objectives (Harley & Cornelissen, 2022; O'Kane et al., 2021; Pratt et al., 2022). Yet fostering trustworthiness in DCS research also brings unique tensions for researchers, which we explain next.

Tensions in Fostering Trustworthiness In DCS Research

We highlight recurring tensions in how DCS researchers conduct and disclose their analytical moves—and how these choices shape what readers can reasonably treat as trustworthy. We surfaced these tensions by drawing on (1) a methodological review of studies using topic modeling, one

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of the most commonly employed DCS tools in IS, management, and organizational research, and (2) interviews with authors who have published DCS research in these domains.[4] The review revealed wide divergence in *how* choices during data work, model building, and validation are justified and disclosed; the interviews helped us understand *why* those choices were made and *why* some were emphasized in published articles while others were not. More details on this review and on the interviews are available in the transparency materials.

We uncovered three recurring trustworthiness tensions: (1) prioritizing objective versus customized data work, (2) seeking stability versus variance in model building, and (3) benchmarking against ground truth versus accounting for interpretations in validation. For each tension, we explain why it arises in DCS research and how either side can be trustworthy. Table 1 summarizes these tensions; the subsections below elaborate on the tensions and motivate the framework that follows in the next section.

Table 1. Trustworthiness Tensions in DCS Research

Tension	Explanation	Salience in DCS research
Data work: Prioritizing objectivity versus customization	*Objectivity* can foster trustworthiness because it constrains researchers' discretion, so surfaced patterns can be attributable to the data, and the path from corpus to model can be auditable via concise, standard descriptions. *Customization* can foster trustworthiness because it enables giving voice to the data, and thereby the phenomenon under study, and helps to explicate rare expressions that might otherwise be missed, improving semantic fidelity and producing interpretable categories supported by snippets from the data.	DCS tools are often used on unstructured textual data; sampling and preparation tasks (e.g., tokenization, removal of stopwords, lemmatization or stemming, constructing *n*-grams, weighting) are more complex and substantively shape downstream categories and dimensions.
Model building: Seeking stability versus variance	*Stability* can foster trustworthiness because researchers can claim that patterns survive reasonable perturbations and demonstrate targeted sensitivity checks with focused evidence for evaluation. *Variance* can foster trustworthiness because researchers can leverage sensitivity as a signal to surface informative alternatives and document what iterations taught about the phenomenon with concrete examples from raw data.	DCS tools can be sensitive to different choices made during data work and model building; models may change with seemingly minor changes and slight parameter tuning; running the same algorithm with the same configurations may still differ because the models may represent locally optimal solutions.
Validation: Benchmarking against ground truth versus accounting for interpretations	*Benchmarking against ground truth* can foster trustworthiness by validating the extent to which labels and subsequent patterns represent the reality of the world represented by the data, including through independent labeling and intercoder reliability or comparison to accepted theoretical frameworks, established methods, accepted metrics, or industry benchmarks. *Accounting for interpretations* can foster trustworthiness by showing that the interpretations by the researcher make sense and can be explained, treating surfaced categories as first-order codes, explaining coding procedures, visualizing data structures, and moving back and forth between model results and the original data to assess fit.	There is often a lack of ground truth in DCS research, and DCS tools are unsupervised, opaque, and unstable; establishing contribution and rigor hinges on how models are validated.

[4] We conducted the methodological review and author interviews as part of a bigger project; hence, our review and interviews included researchers employing and designing DCS tools for constructing as well as testing theories. For the purpose of this study, our focus is limited to only those studies where researchers employed DCS tools interpretively for constructing theories, i.e., studies within the CTC community.

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Data Work: Prioritizing Objectivity vs. Customization

DCS researchers spend considerable time collecting and preprocessing texts into a machine-readable format, an activity we refer to as data work. Unlike traditional statistical analyses on structured data (e.g., regression), DCS work typically begins with unstructured texts, where guidance is abstract. Researchers' choices (e.g., sampling, filtering words, and normalizing) materially shape the categories and dimensions that are later surfaced. This is because "seemingly minor changes such as preprocessing or changes in the underlying data" may change the model outputs (Grimmer et al., 2022, p. 227). The core tension is whether to foster trustworthiness by aiming for *objectivity*—constraining discretionary preprocessing and relying on standard rules and thresholds—or by pursuing *customization*—tailoring preprocessing to the phenomenon to preserve semantic fidelity.

On the one hand, some DCS researchers choose objectivity as a path to trustworthiness by limiting the amount of processing they do, following commonly accepted data preparation steps (e.g., only removing "common" stop words), and utilizing computational metrics to objectify decisions related to data work (e.g., cutoff points based on frequency). As explained by one interviewee, such practices can facilitate objectivity while avoiding researcher biases.

> *Of course, you can run different topic models with different sets of stop words. It's something that I think we have also done to some extent when running those different topic models. But … we rather didn't include too many [or] didn't exclude too many words so that we did not have a bias in the results.* (Interviewee 1)

Scholars also minimize upfront data preprocessing and prefer to remove non-meaningful categories only after having built the model, thereby "letting the data speak for themselves," as several interviewees explained. Again, the belief is that by minimizing researcher intervention and therefore bias, results become more objective. For example, one interviewee explained how reviewers questioned the objectivity of the results:

> *We were addressing this potential concern of "we weren't being objective as researchers," which is kind of like a hard thing to hear, right, like your reviewers are essentially challenging your integrity, like, "you lied to us in your initial submission."* (Interviewee 2)

On the other hand, we also learned that some DCS researchers choose customization in their data work as a path to trustworthiness because they want to engage more closely with the data. While closer engagement with data may introduce bias and perhaps make the results less objective, it does help researchers to justify their choices in relation to the specific research context. For example, one interviewee explained how their data preparation choices were based on an understanding of the data and the context:

> *It's a choice informed by how you understand the phenomenon, how you understand the data. I think this [bias-free] picture is, from my perspective, a bit naive; it's just a choice that you make, and what I see as my duty as a researcher is to make this choice transparent and available to the others for assessment.* (Interviewee 3)

In making these choices, however, DCS researchers may face challenges in predicting how choices influence model outcomes, causing them to engage in creative experimentation rather than following rigorous templates. For example, one interviewee explained how they iterated with different preprocessing steps to see how these choices influenced the emerging categories:

> *I spent a lot of time choosing which words to kick out and which words to keep in, trying to keep in as much as possible, but running some different models to see what keeping a very common word might do. So, in that preprocessing stage is where I think a lot of the choices I make—and it's hard to say that there's like a hard and fast rule on this right. I think this is where we make lots of choices as researchers in our studies.* (Interviewee 2)

Overall, DCS researchers are often torn between two competing approaches to fostering trustworthiness in their data work: either through the path of objectivity or through customization. Both approaches are valid. Objectivity facilitates trustworthiness because it constrains the researchers' discretion; hence, surfaced patterns can be attributed to the data (conduct), and researchers can thus make the path from corpus to model auditable via concise, standard descriptions (disclosure). At the same time, customization also facilitates trustworthiness because it allows for explicating rare expressions in data that might otherwise be missed, improving semantic fidelity (conduct) and producing interpretable categories supported by snippets from the data (disclosure). Absent guidance, authors struggle in deciding which path to prefer and how to convincingly report their choices.

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Model Building: Seeking Stability vs. Variance

DCS researchers face a central dilemma in model building: seeking *stability* versus *variance*. Again, this challenge is prominent in DCS research because model outputs tend to be very sensitive to the choices of algorithms and parameter tuning (Belford et al., 2018; Grimmer et al., 2022). Given this sensitivity, deciding whether to foreground robustness checks or use variability as an exploratory probe is, in itself, a consequential choice for fostering trustworthiness.

DCS researchers may pursue stability in model building because they may conceive sensitivity as a bug that needs fixing (Schooler, 2014), and doing so creates a path to trustworthiness. Thus, researchers may actively ascertain how the choices they make in building their models influence the model output through performing sensitivity tests. Examples of sensitivity analyses include varying parameter values or weighting schemes (e.g., Haans, 2019; Lowry et al., 2013), using left-out samples or different data sources (e.g., Geva et al., 2019; Gong et al., 2018), changing the modeling algorithms (e.g., Bao & Datta, 2014; Larsen & Bong, 2016), or running the same model with multiple random seeds (Belford et al., 2018). Performing such sensitivity tests allows researchers to claim that their model output is not sensitive to specific choices they made in building the model. Following this standard practice, DCS researchers also increasingly rely on and report sensitivity analyses to demonstrate that the model and the results are *stable* across choices that have been made in the different research phases. This rationale reflects the thoughts of several interviewees, as expressed in the following quote:

> *And usually what we do is ... we try [to see] how sensitive this choice is. We can't always do this for all the parameters, of course. But as often as we can, this is what we try to do and to see whether, how much this parameter weighs in shaping the results.* (Interviewee 3)

DCS tools are not only sensitive to choices but are also inherently sensitive, because running the same algorithm using the same configurations on the same data could lead to different results (Belford et al., 2018; Grimmer et al., 2022). In response, scholars have responded by developing more stable tools—such as Yang and Subramanyam's (2023) *stable topic model*—to mitigate instability that can produce inconsistent, irreproducible results.

Alternatively, a model's sensitivity may also help in demonstrating trustworthiness, as opposed to pointing to a bug. For example, one interviewee explained how, while experimenting with different configurations, some intermediate models enabled them to capture an event that was only minimally reported on in the documents that they were analyzing but was potentially relevant for improving the understanding of their specific phenomenon. These models and results were not considered stable enough, as using alternative configurations would make the specific category disappear. But this example illustrates how experimenting with different configurations can enable researchers to explore alternative explanations and categorizations, some of which might be insightful. Another interviewee highlighted how it is then up to the researcher to interpret the explanatory value of alternative models and reflect on their choices:

> *To convince me also as a reviewer, people should, for example, mention that they did some iterations, you know, that they did not just use the first topic model that the algorithm gave them. And actually, if they report about some challenges that they experienced while doing it, I would trust them more than if they just left out the challenges.* (Interviewee 1)

Overall, sensitivity appears to be a double-edged sword for DCS researchers. Stability fosters trustworthiness by demonstrating robustness (conduct) and presenting focused sensitivity evidence (disclosure). Variance fosters trustworthiness by using perturbations to explore alternative explanations (conduct) and documenting interpretive iterations with concrete text-level examples (disclosure). Due to the lack of guidance specific to DCS research, scholars tend to be conflicted in deciding which path to trustworthiness they should choose.

Validation: Benchmarking Against Ground Truth vs. Accounting for Interpretations

Establishing validity is one of the most important aspects of DCS research, given the unsupervised approaches to model building: "the primary justification for using text analysis methods is empirical" (Grimmer et al., 2022, p. 30). Accordingly, establishing a theoretical contribution becomes very much a question of fostering trustworthiness in how the models, through which the patterns are surfaced, are validated. DCS researchers need to make a choice in terms of what kind of validation practices to prioritize (DiMaggio et al., 2013), where some practices are driven by the pursuit of *ground truth to benchmark against* and others by the pursuit of researchers' *interpretations to account for*.

Many DCS researchers demonstrate the internal validity of their models by focusing on how well the model captures distinct categories or dimensions and to what extent the researcher's interpretation is fitting. Researchers typically examine the surfaced categories based on the top words and

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assign a label to each category in an attempt to validate the usefulness of categories in identifying distinct concepts: "I think a common practice across all topic modeling work is coming up with labels for topics" (Interviewee 4). When driven by the pursuit of ground truth, researchers tend to focus on validating the extent to which those labels and subsequent patterns actually represent the reality of the phenomenon. They can do so in different ways. For example, researchers may come up with labels *independently* and subsequently compare them to minimize subjectivity. The idea is that minimizing interpretative variety aids in facilitating trustworthiness by identifying patterns that can be considered a form of ground truth. For example, one interviewee explained how they were pushed by reviewers to consider intercoder reliability:

> *... this definitely came from reviewers. They wanted us to confirm this [our interpretation]. This was not our idea. ... And I can see why they wanted this because, indeed, there can be a lot of, you know, subjectivity in it.* (Interviewee 5)

Another validation approach in the pursuit of ground truth is to compare model output with accepted theoretical frameworks and measures, as they have already been proven and observed in similar studies. Similarly, researchers may compare their results to those of studies that apply more established methods or accepted metrics, or to industry benchmarks. Such external validation practices help to illustrate that emerging patterns represent some ground truth. However, interviewees also acknowledged that these forms of validation may be very difficult in the case of DCS tools, due to their unsupervised nature and the inherent lack of ground truth:

> *We kind of used the deductive approach. So now we had a theory and tried to map topics to this theory, but then you have kind of a contradiction, actually, right? Because topic modeling is unsupervised and inductive, and [the approach is deductive]. You see what I'm talking about. And that's difficult to resolve, right?* (Interviewee 1)

Thus, DCS researchers may also lean towards fostering trustworthiness through accounting for their interpretations. The focus in this case is less on showing that patterns reflect some ground truth than on showing that the interpretations by the researcher make sense and that the researcher can explain the interpretations. Oftentimes, researchers treat categories resulting from their models as something akin to first-order codes in grounded theorizing and ultimately exclude or combine categories under more general second-order or aggregate themes. In such cases, researchers strive to demonstrate trustworthiness by explaining their coding procedures and visualizing clear data structures. For example, one interviewee noted how trustworthiness expectations in some CTC studies may be similar to those in qualitative studies:

> *I would tend to think that a [qualitative research] paper that's doing a grounded theory or a Gioia technique, you absolutely need to show your data structure. But you know, if you think about [example DCS paper], you know their first-order codes were topics.* (Interviewee 4)

> *If you're just doing a study where you're isolating one topic, then you know it is helpful to the reader to at least have an appendix to get a sense of what the other topics were, so that they could potentially argue with your selection of that topic versus another.* (Interviewee 4)

The challenge for DCS researchers then is that it may not always be clear why or how patterns surface as a result of applying certain algorithms and whether the researcher's interpretations actually align with the assumptions built into those algorithms. As a consequence, DCS researchers may find themselves constantly going back and forth between model results and the original, underlying data. For example, one interviewee argued that it is crucial to make sure that the interpretation of topics aligns with what is present in the data:

> *Assuming you're not just like throwing an algorithm at something, you still have to be very careful with how you're choosing your topics and how you're validating your model and going back to your text data. ... How does it map? Does it make sense? And then once you do those checks, which is still considerable work, it acts as another point to validate.* (Interviewee 2)

Thus, both benchmarking against ground truth and accounting for interpretations can ground validity and thereby foster trustworthiness while justifying different kinds of claims and evidence. Absent guidance, authors sometimes try to do everything, combining approaches that pull in opposite directions and leaving DCS researchers, reviewers, and editors unsure which evidentiary logic should carry the most weight.

Synthesis of Tensions

A key thread that runs across data work, model building, and validation is that DCS research cannot follow a generic approach in fostering trustworthiness. This is because DCS research blurs the boundaries between quantitative (including

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computational) and qualitative approaches; however, it is neither qualitative nor quantitative nor even a mixed method (Berente et al., 2019; Lindberg, 2020). As one of our interviewees explained: "I feel that's the risk of using [DCS tools], which you can use in both qual and quant. And if you then don't get people [in the review process] who are open-minded to something that's not either perfectly qual or perfectly quant, ... as I've experienced, this can be really tough" (Interviewee 6). Prioritizing objective versus customized preprocessing in data work, seeking stability versus variance in model building, and benchmarking against ground truth versus accounting for interpretations in validation can each foster trustworthiness, but for different reasons. These tensions motivate our framework, which specifies how research objectives can inform those reasons.

A Framework for Fostering Trustworthiness in DCS Research

We propose a framework to help DCS researchers navigate trustworthiness tensions. We clarify trustworthiness expectations across two research design choices that set expectations for burden of proof and minimum disclosures. The first choice concerns whether DCS researchers give primacy to either *theoretical lexicons* (i.e., starting with theory) or *practice lexicons* (i.e., starting with the phenomenon), creating two pathways of DCS work. The second choice concerns whether the *content of texts* (i.e., meaning-bearing distinctions) or the *structure of the corpus* (i.e., attributes of the text corpus) carries the theoretical load. In our framework below (see Figure 1), we organize expectations for conduct and disclosure across the three phases discussed in Table 1.

We use select DCS studies from IS, management, and organizational research—including Hacker et al. (2020), Hannigan et al. (2022), Karanović et al. (2021), Croidieu and Kim (2018), Giorgi et al. (2019), and Miranda et al. (2022b)—as illustrative examples to explain our framework. Table 2 summarizes the trustworthiness moves by these studies. The first column locates each study within the two dimensions of our framework, the second column documents how the study uses DCS tools to explicate the focal phenomenon, the third column summarizes the trustworthiness moves the study performs well, and the final column highlights additional trustworthiness moves that our framework would recommend.

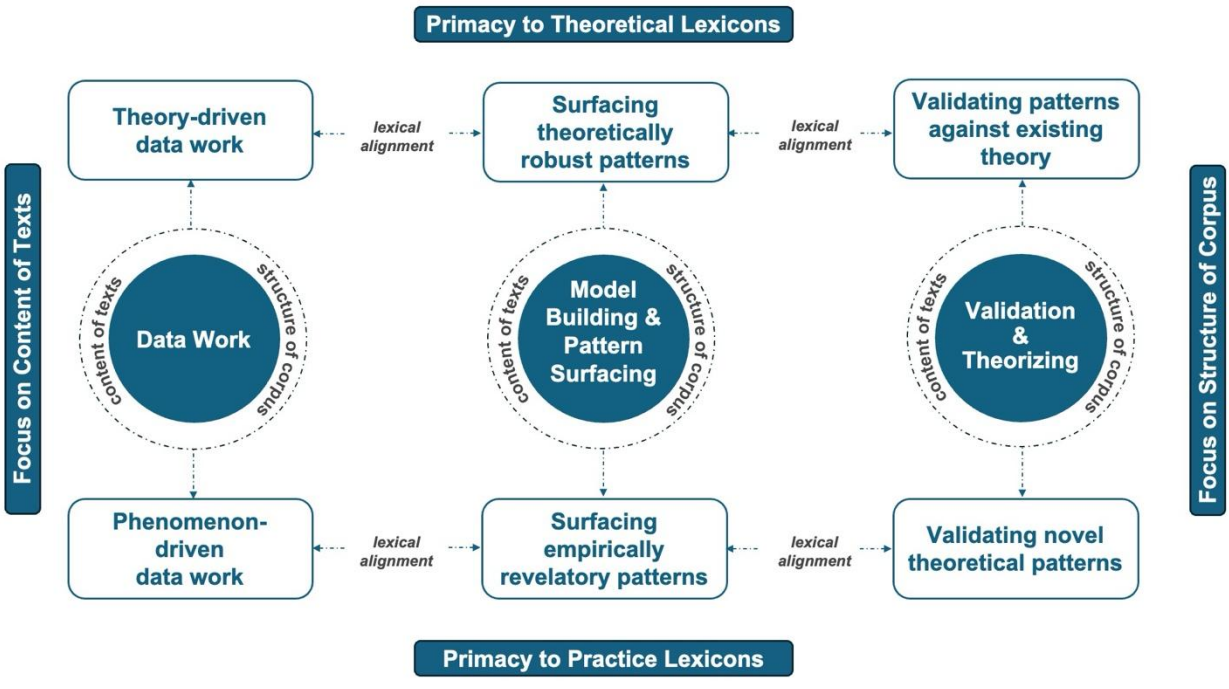


Figure 1. Framework for Fostering Trustworthiness in DCS Research

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Table 2. Trustworthiness Moves in the Illustrative Studies

Exemplar study & design choices	Focal phenomenon and role of DCS tools	Trustworthiness moves already done well by exemplar study	Additional trustworthiness moves recommended based on framework
Hacker et al. (2020) *Primacy:* Theoretical lexicons *Focus:* Content of texts	Examine how web conferencing systems (WCS) were used during COVID-19; use topic modeling on tweets to surface topics interpreted as WCS affordances.	- Using affordance theory to script sampling; Targeted cleaning while preserving lexical detail to distinguish affordances. - Choosing algorithms that have proven to do well on tweets. Exploring alternative numbers of topics. - Facilitating face validity of topics interpreted as affordances.	- Justifying preparation steps by more explicitly relating them to the theory of affordances and the focus on semantic meaning. - Treating parameters beyond the number of topics and reporting and justifying them in relation to their ability to surface affordances. - Offering a theory-grounded rationale for dropped models and topics, showing that their exclusion does not undercut claims about WCS affordances.
Hannigan et al. (2022) *Primacy:* Theoretical lexicons *Focus:* Structure of corpus	Map how culture manifests in an entrepreneurial ecosystem; apply topic modeling to social-media talk to analyze cultural patterns and trajectories.	- Using cultural theory to guide sampling focused on capturing ecosystem talk during a formative period. Preprocessing with the aim to capture associations and trajectories. - Choosing an algorithm more suited than default choice, for documents with a small number of words. - Strengthening internal validity by having observers independently code topics and reconcile to consensus.	- Specifying preprocessing beyond lemmatization and linking it explicitly to the structural mapping aim. - Reporting parameter experimentation and showing attribute invariance under reasonable perturbations. - Tying structural shifts more explicitly to recognizable field or ecosystem episodes.
Karanović et al. (2021) *Primacy:* Theoretical lexicons *Focus:* Structure of corpus	Study how resource dependence conditions platform workers' responses across regulatory regimes; use topic modeling on forum posts to compare patterns across cities/regimes.	- Using resource dependence theory to anchor sampling choices and theory-driven stratification. - Choosing an algorithm enabling structural comparison, with city and date as prevalence covariates so topic prevalence is directly tied to regulatory context. - Assessing reliability of topic interpretations by recruiting independent coders. Bolstering external validity through triangulation of multiple data sources on workers' responses across regulatory structures over time.	- Considering preprocessing steps beyond tool defaults, justifying them in terms of comparability vs interpretability across cities and time. - Treating parameters beyond the number of topics and reporting and justifying them depending on their ability to surface structural resource differences. - Emphasizing robustness of structural findings by showing city/date effects and whether key patterns persist across nearby configurations.

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Croidieu and Kim (2018) *Primacy:* Practice lexicons *Focus:* Content of texts	Trace how amateur and lay expertise became legitimized in the amateur radio field; use topic modeling to surface and follow shifting articulations of amateur.	- Using phenomenon-driven sampling through dropping peripheral material, relating different sources, and sequencing episodes to track shifts in amateur. Preprocessing focused on retaining meaning. - Choosing the algorithm that addresses the polysemy of "amateur" and has proven useful for analyzing historic texts. Exploring different configurations of parameters and disclosing choices leading to the final model. - Designing and reporting on coding procedure and visualizing code structure. Providing quotes to substantiate arguments and enable interpretation.	- Making explicit how the phenomenon motivated specific preprocessing steps and split/merge decisions. - Building on topic stability checks by reflecting on the revelatory value of excluded topics and why they were set aside.
Giorgi et al. (2019) *Primacy:* Practice lexicons *Focus:* Content of texts	Reconstruct how firms responded to the 1966 Automotive Safety Act; use topic modeling to organize heterogeneous safety-related materials.	- Assembling a heterogeneous corpus on firms' responses to the Safety Act to build a revelatory case grounded in practice. - Choosing an algorithm that addresses polysemy in historical contexts. Reporting on the specific tool used. Exploring alternative numbers of topics. - Triangulating qualitative and quantitative analysis. Designing a coding process based on independent coding. Anchoring topics in genre-specific texts with quotes.	- Justifying preprocessing steps in relation to claims about safety discourse rather than positioning it as a "standard" procedure. - Treating parameters beyond the number of topics and reporting and justifying them depending on their ability to reconstruct firms' responses. - Exploring if alternative choices lead to alternative frames and assessing their explanatory value.
Miranda et al. (2022b) *Primacy:* Practice lexicons *Focus:* Structure of corpus	Examine how field-level blockchain discourse evolves in diversity and coherence; use topic modeling and diversity/coherence measures on a field-level corpus.	- Constructing a corpus of blockchain discourse across heterogeneous outlets and actors through phenomenon-driven sampling. Data preparation focused on retaining meaning while improving structural comparison. - Using algorithms that support surfacing meaning structures and computing diversity/coherence over time. Reporting on the tool and specific parameter settings leading to the final model. - Supporting validation with synchronic/diachronic visuals and quotes.	- Making lexicon provenance explicit: clarify how they ensured they did not miss any sources beyond the theory-informed sampling. - Highlighting the revelatory value of the chosen model vis-à-vis alternatives.

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Giving Primacy to Theoretical Lexicons

When researchers give primacy to theoretical lexicons, their research questions are anchored in a prior theory, and the ensuing choices across phases are guided by what that theory renders salient. The aim is to unpack how a particular theoretical lens illuminates a focal phenomenon (see Table 2): for example, how cultural theory can explain entrepreneurial ecosystems (Hannigan et al., 2022), how resource dependence theory can explain the gig economy (Karanović et al., 2021), or how affordance theory can explain the use of web-conferencing systems (WCS) during the COVID-19 pandemic (Hacker et al., 2020). Yet this aim can be achieved by focusing either on the content of texts or on the structure of the entire corpus. Hacker et al. (2020) privileged content because category meanings bear the theoretical load (i.e., categories representing affordances), whereas Karanović et al. (2021) and, to a large extent, Hannigan et al. (2022) privileged structure in their respective focus on distributions across regulatory contexts and associations and trajectories in cultural discourse.

In the following, we articulate how researchers can facilitate trustworthiness when giving primacy to theoretical lexicons through theory-driven data work, surfacing theoretically robust patterns, and validating patterns against existing theory. We also explain how lexical alignment can be achieved through iterative sensemaking.

Theory-Driven Data Work

When DCS researchers give primacy to theoretical lexicons, data work reflects a theory-driven orientation. Because other studies may already have analyzed related phenomena under similar theoretical framings, DCS researchers can use them as starting points for maneuvering theory-consistent data work choices, beginning with sampling.

In theory-driven sampling, theory-anchored research questions implicitly or explicitly script analytical moves in data work: dropping sources that do not speak to the theorized mechanism, relating focal actors and contexts into a purposive sampling frame, and sequencing observations to align with expected dynamics. For instance, in asking how culture manifests in an entrepreneurial ecosystem, Hannigan et al. (2022) defined the unit of analysis as ecosystem members' public talk (posts on Twitter, now X). They excluded non-ecosystem discourse and bound the observation window to the ecosystem's formative period so that cultural fluctuations could be observed. In asking how resource dependence conditions platform workers' responses across regulatory regimes, Karanović et al. (2021) defined strata by the city-level regulatory context and sample posts from uberpeople.net. They argued that platform "workers' responses to new organizing solutions are likely to vary between regulatory structures" (p. 6). Accordingly, they bound observations by jurisdiction and time to capture variation aligned with theorized dependencies. They also drew on prior work (e.g., Tzur, 2019) to justify the salience of regulation for transport network companies and motivate forums as data sources that have proven to be "valuable sources of insight into workers' experiences" (Karanović et al., 2021, p. 2). This helps explain why they did not include venues and threads unrelated to ride-hailing work. In asking which WCS affordances were actualized during COVID-19, Hacker et al. (2020) sampled tweets about WCS use. Affordance theory guided their data work, while they remained open to emergent themes. Consistent with the theoretical focus on WCS affordances during the crisis, they dropped off-topic chatter and bound the observation window to the crisis period. In each of these studies, the guiding theory anchored the sampling frame and justified early dropping, relating, and sequencing moves as part of theory-driven data work.

Theory also steers data preparation as active categorization (see Grodal et al., 2021). To an extent, this is visible in how scholars prepare data in line with some agreed-upon standards and practices associated with the chosen theory. In practice, this entails various moves in processing data that dictate the categories that will eventually surface, such as filtering non-substantive lexical material (e.g., stopwords, usernames/URLs, numbers, punctuation), and normalizing lexical variants (e.g., lemmatization, synonym handling, and preserving salient *n*-grams). For example, Hannigan et al. (2022) specified what tool they used for lemmatizing tweets, and Hacker et al. (2020) reported removing usernames, URLs, numbers, and punctuation, applying bi-gram tokenization, extending stoplists with domain-uninformative terms, and lemmatizing. Across these cases, the guiding theory shaped which preparation choices were emphasized and why, and the authors explicitly reported on theory-motivated steps so that readers could see how early decisions set up downstream DCS work.

The initial research questions and theoretical orientation also shape data work by clarifying whether category content is expected to carry the theoretical load or whether structure is central. When the study is content-focused, sampling foregrounds texts that narrate concrete actions tied to focal constructs, and data preparation preserves semantic distinctiveness. For example, given that the theory of affordances is about how people use technology, Hacker et al.'s (2020) data work centered on the content of texts. They applied targeted cleaning (e.g., removing usernames) while retaining the lexical material needed to differentiate affordance meanings. When the structure of the corpus is in focus—e.g., the theory targets associations among categories or documents—data work prioritizes the structural signal and comparability. Researchers sample in a balanced manner across theory-implicated strata (e.g., cities, periods, roles) and capture

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rich metadata to support later relating and sequencing. Researchers may also stick to tool defaults to keep preprocessing uniform across strata (e.g., the same language filters and stop-word lists) (Karanović et al., 2021) and avoid content-motivated pruning that differentially thins or collapses groups or time periods.

Using our framework, the illustrative studies' choices in theory-driven data work can be complemented with certain analytical moves and disclosures that strengthen trustworthiness. For instance, Hannigan et al. (2022) could have specified whether any preparation beyond lemmatization was undertaken and why, so that readers can see how preprocessing aligns with their theoretical aims. Karanović et al. (2021) could have considered steps beyond tool defaults. For example, they could have explicitly stated whether they used stemming or lemmatization and justified that choice given their aim of comparing structural variation across regulatory contexts (where stronger normalization aids comparability but may reduce interpretability). All studies could have also reflected on what constitutes data and made their decisions about data more explicit to their readers. For this, DCS researchers can draw inspiration from some studies that use DCS tools to operationalize constructs. For instance, Safadi et al.'s (2021) study on Stack Exchange is a useful case for theory-driven data work in clarifying what counts as data: They treated the contribution text as the data (one post—question, answer, or comment—constitutes a document) and explicitly rejected member-selected tags as content proxies because tags appear only on questions, reflect author idiosyncrasies, vary by community, and can be auto-suggested. Such clarifications are important for addressing the risk that the practices a given study follows are assumed to be "best practices" even when they are not meant to be best practices by the authors who introduce them. To mitigate this risk, we encourage researchers to disclose and justify preparation steps in relation to the specific theory they are focusing on.

Surfacing Theoretically Robust Patterns

After theory-driven data work, model building turns on two decisions: (1) the choice of algorithmic family and (2) parameter settings. The choice of the number of categories governs splitting versus merging of categories, priors, and regularization shape how sharply categories are expressed, initialization and random seeds test whether salient patterns reappear, and assignment thresholds or prevalence cutoffs implement disciplined dropping of borderline memberships that would otherwise blur patterns. Through these choices, researchers identify patterns that are analytically robust and interpretively meaningful for the chosen theory. Trustworthiness rests on showing reasonable alternatives around both decisions (algorithms and parameters) and what would (or would not) change under nearby specifications. We elaborate on these choices below.

First, giving primacy to theoretical lexicons demands selecting algorithms and configuration norms from prior studies with similar contexts, theories, and data so that results are comparable and knowledge can be accumulated. For example, Hacker et al. (2020) used the MineMyText online tool, which relies on the latent Dirichlet allocation (LDA) (see Blei et al., 2003), because "it is well-researched from both the technical and the application perspectives, [including] its use in analyzing tweets" (p. 566). Yet, researchers still ought to be mindful of the underlying assumptions inscribed in those algorithms. Trustworthiness is facilitated by carefully explaining how the chosen algorithms and underlying assumptions fit the research context and considering potential deviations when the data and theory require it. For example, Hannigan et al. (2022) explained why they rely on bi-term analysis instead of the standard LDA: "When each document (i.e., a tweet, in our current case) has a small number of words, LDA is not optimal. The distribution of topics across the documents was too sparse, making estimation of the LDA parameters for a document particularly difficult" (p. 7). Karanović et al. (2021) adopted the structural topic model (STM) (see Roberts et al, 2014), so topic prevalence can be related to city- and date-level metadata during estimation, matching a resource-dependence argument about regulatory conditioning. These examples illustrate why researchers should explain the reason for choosing a specific algorithm for the intended purpose, explain the core assumptions that come with that algorithm, and indicate how reasonable alternatives would be expected to change (or not) the patterns central to the theory-led questions.

Second, parameters are the levers that enact active categorization. Parameter choices are not mere technicalities; they are concrete means by which theory-driven research questions are rendered into model behavior. Researchers can start near established defaults, then explore nearby settings to select a configuration that is both theoretically meaningful and empirically robust. For example, Hacker et al. (2020) qualitatively compared models with 20, 30, 50, and 60 topics and chose 50 because it improved differentiation without adding empty complexity. While the other illustrative papers also experimented with the number of topics, they could have explained how this may have influenced the patterns observed and if this also led to patterns that could not be explained with the help of the existing theory.

When the study is content-focused, researchers can choose settings that enable more precise category splitting, with robustness centering on whether content labels persist under

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nearby choices of number of categories, priors, or seeds. By contrast, when a structural attribute carries the theoretical load, researchers can favor algorithms and parameter settings that emphasize relating categories and documents to covariates and sequencing them over time or interaction (e.g., STM to relate prevalence to city/date). Researchers in such cases should be cautious about how parameterization affects prevalence or co-occurrence in the structure of the corpus. In short, in terms of fostering trustworthiness, theory-led questions determine whether model building primarily entails analytical moves to refine meanings (focus on content) or to preserve and interrogate attributes (focus on structure).

Viewed through our framework, the illustrative studies could have experimented (and disclosed experimentation) with parameter configurations beyond the most obvious ones. For instance, in the case of topic modeling, in addition to the number of topics, concentration priors (α, β) materially shape granularity and sparsity, and should be reported and justified to enable knowledge accumulation. DCS researchers giving primacy to theoretical lexicons could clearly link their justification to the theory that guides their analysis. For example, in the case of Hacker et al. (2020), had they chosen a lower value for β, they may have uncovered more fine-grained topics that they could then have interpreted through an affordances lens. Similarly, Karanović et al. (2021) could have reported that the direction and magnitude of city/date effects on topic prevalence persisted across nearby configurations. Thus, DCS researchers can draw inspiration from scholars who use DCS tools for measuring deductively developed constructs, e.g., how Kaplan and Vakili (2015) considered different values of β, and how Mousavi and Gu (2024) experimented with different algorithms, including BERT, LSTM, and a dictionary method, before settling for a fine-tuned BERT for the rest of their analysis. These analytical moves (conduct and disclosure) help other researchers to interpret parameters in studies relying on similar theories, thereby fostering trustworthiness.

Validating Patterns Against Existing Theory

After surfacing patterns through model building, researchers need to validate the emergent patterns against the chosen theories. Validation rests on two pillars: internal and external validity. First, internal validity under the primacy of theoretical lexicons establishes that the category system faithfully instantiates the guiding theoretical constructs (or their theory-relevant attributes) and that the resulting patterns are not merely artifacts of modeling choices. A first step is face validity—Do outputs look reasonable in the language of the theory? This typically includes visualizing categories (top words and top documents) with provisional, theory-anchored labels. Intersubjective interpretability can be strengthened with multiple coders using a brief, theory-informed codebook and reconciliation. For example, in Hannigan et al. (2022), three observers independently coded topics and resolved differences based on consensus.

Face validity is necessary but not sufficient, especially when the focus is on the content of texts. In this case, authors can show label persistence and exemplar stability under a varying number of categories, seeds/initializations, and priors. Internal validation can also make explicit the active categorization at stake: how labels reflect splitting/merging decisions, what (if anything) is dropped as noise, and how categories are related. Such computational-interpretive triangulation and robustness testing can clarify how far theory-led interpretations travel and support cumulative knowledge. When the theory targets a structural attribute, face validity can be paired with checks to confirm that structural metrics are invariant to reasonable perturbations and denominator integrity is preserved.

Second, external validation of DCS output under the primacy of theoretical lexicons strengthens the claim that theory-led interpretations travel beyond the modeling setup. One route is triangulation with independent data or methods. For example, Karanović et al. (2021) triangulated different data sources "to compare workers' responses under different regulatory structures and analyse these responses over time" (p. 2). In a similar spirit, studies can complement model outputs with category-sequencing and event anchoring—showing that local discourse reacts to field-level developments (e.g., salient AI milestones or policy shifts) in theoretically coherent ways. Following our framework, Hannigan et al. (2022) could have made this link explicit by demonstrating that the model's cultural associations co-moved with recognizable inflection points in the broader AI landscape.

The other route is construct alignment: comparing surfaced categories and dimensions to preexisting theoretical frameworks or taxonomies to assess convergent/discriminant validity and the value-add of DCS (e.g., new subcategories or clearer boundaries). For example, after completing a literature review and under the guidance of the theory of technology affordances in organizations, Hacker et al. (2020) decided to drop topics that were difficult to interpret or did not align with the chosen theoretical framework. When such dropping occurs, authors can justify it in theoretical terms and show that it does not undercut the claims they make later.

Lexical Alignment Through Iterative Sensemaking

So far, we have explained how researchers employing DCS tools in constructing theories with primacy given to theoretical lexicons can continue to maintain proximity to theoretical lexicons within each research phase. But we have also argued that researchers align their theoretical and practice lexicons

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across phases iteratively. These iterations can render theory-led questions analytically tractable. Our framework asks authors to report a concise iteration trace: (1) the theoretical motivation for returning to an earlier phase, (2) the specific move enacted (e.g., added domain stopwords; changed number of categories/priors), (3) the observed effect on content (labels/exemplars) or structure (prevalence/ associations/ trajectories), and (4) why the post-iteration state better instantiates the theory. Making these cycles explicit shows not only where the analysis lands but also demonstrates, step by step, how theory and practice lexicons are brought into correspondence.

Giving Primacy to Practice Lexicons

An alternative research design choice in DCS research is giving primacy to practice lexicons. When researchers give primacy to practice lexicons, their research questions are anchored in a focal phenomenon and the vocabularies through which actors themselves make it intelligible. The ensuing choices across phases are guided by what the phenomenon's vernacular renders salient. The aim is to surface meanings and mechanisms inherent in practice and only subsequently stabilize them in dialogue with theory (see Table 2): for example, how amateurs' discourse comes to be recognized as lay expertise in an emerging field (Croidieu & Kim, 2018), how firms and other actors reshape the cultural terms through which legal issues are debated (Giorgi et al., 2019), or how blockchain discourse becomes simultaneously diverse and coherent across multiple discursive fields (Miranda et al., 2022b). However, this aim can be achieved by focusing either on the content of texts or on the structure of the corpus. In the following, we articulate how DCS researchers giving primacy to practice lexicons can foster trustworthiness through phenomenon-driven data work—surfacing empirically revelatory patterns and validating novel theoretical patterns—followed by an explanation of lexical alignment through iterative sensemaking.

Phenomenon-Driven Data Work

When DCS researchers give primacy to practice lexicons, their data work ought to be phenomenon driven. Researchers sample the corpus by iteratively unpacking the focal phenomenon: They select sources that speak to the phenomenon under study and relate actors, venues, and arenas into a purposive frame that captures its contours. For instance, Croidieu and Kim (2018) built a historical, multi-actor archive around the rise of amateur radio, relating regulators, associations, and media to situate the practice in its field, and sequencing documents across key episodes to observe shifts in how "amateur" was articulated. Giorgi et al. (2019) applied topic modeling to a wide range of materials on firms' responses to the 1966 Automotive Safety Act to assemble a "revelatory case in which the processes of theoretical interest were 'transparently observable' and could be thoroughly reconstructed" (p. 807). Likewise, Miranda et al. (2022b) constructed a field-level corpus to study discursive dynamics, relating heterogeneous outlets and voices that constitute the arena and sequencing materials over periods salient to field evolution. In each case, the phenomenon anchors the sampling frame and justifies these early moves as part of phenomenon-driven data work.

Data preparation and preprocessing are acts of sensemaking rather than applications of standards. In this pathway, active categorization involves a set of ex ante moves—relating, sequencing, and merging—that renders practice-rooted distinctions legible before any modeling. Because theoretical lexicons have not yet been stabilized, preparation emphasizes preserving lexical richness while documenting the micro-moves that shape what can eventually be surfaced. What counts as noise versus meaningful data is grounded in the researchers' understanding of the phenomenon, which guides dropping and merging decisions. In Croidieu and Kim (2018, Appendix p. 1), tracing the meaning of the word "amateur" implies keeping local context windows, identifying field actors to relate voices across episodes, and omitting "words considered to be non-meaningful." In Miranda et al. (2022b), field-level discourse dynamics required retaining semantic units in the form of bi-grams and performing lemmatization, while also sequencing stable time bins so that the later diversity and coherence measures were comparable. Our recommendation is to treat these choices as provisional knowledge claims about the phenomenon: Document each action (what was dropped or kept, how synonyms were merged, which metadata were related, how periods were sequenced), revisit after an initial rendering, and justify adjustments in terms of how they clarify the phenomenon under study.

When content is central—as in Croidieu and Kim (2018) on the shifting senses of amateur and, to a large extent, Giorgi et al. (2019) on safety discourse—DCS researchers engage in sampling that foregrounds texts that narrate concrete meaning-bearing practices and discourses, and in preprocessing that preserves semantic meaning so subsequent category splitting and judicious merging are possible. When structure is central—as in Miranda et al. (2022b), where diversity, coherence, or related field attributes do the explanatory work—data work protects structural signal and comparability. In such cases, DCS researchers relate outlets, periods, and actor types via rich metadata, put sequence observations in revelatory bins aligned with field episodes, and apply cleaning so that structural measures reflect the arena. Across both aims, the phenomenon anchors which

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moves are appropriate.

Using our framework, the illustrative studies could have enhanced trustworthiness by tweaking some analytical moves. For instance, Giorgi et al. (2019) noted that they "followed a standard procedure" (p. 824), citing Croidieu and Kim (2018). Yet the two studies confront different corpora, genres, and phenomena, so a shared "standard" risks obscuring phenomenon-specific needs. Our framework suggests that studies that give primacy to practice lexicons should treat preprocessing as provisional theorizing and document it as such. Concretely, Croidieu and Kim (2018) could have made explicit whether their historical context and polysemy of words warranted their decision to stem words, or if lemmatizing would have been more applicable, given their focus on content and interpretation as well. Giorgi et al. (2019) could have leveraged and justified context-aware stoplists, showing how these moves clarified safety discourse and demonstrating that preprocessing did not undermine dissenting voices. Miranda et al. (2022b) already documented preprocessing and sampling in depth; one further step could have clarified how they ensured that relevant discursive strands outside of their theory-informed sampling were not missed. Overall, researchers operating close to practice lexicons can foster trustworthiness by reflecting on how choices help them understand the phenomenon and describing the iterative sense-making through which these choices are refined so that they avoid taking "standards" for granted.

Surfacing Empirically Revelatory Patterns

When researchers give primacy to the practice lexicons, after phenomenon-driven data work, model building aims to surface revelatory patterns—those that open new concepts or mechanisms and remain credible under reasonable perturbations. Again, two central decisions include algorithms and parameters. First, the choice of algorithmic family functions as a lens to give voice to the phenomenon. For instance, Croidieu and Kim (2018) chose LDA because it distinguishes cases in which a word's meaning shifts by context, thus allowing for capturing polysemy. Giorgi et al. (2019) implemented LDA to render an interpretable discursive architecture and because it enables capturing polysemy. Miranda et al. (2022b) explained that they followed their choice of using LDA to render topics with "multidimensional scaling (MDS), which surfaces underlying meaning structures by positioning related topics next to each other in two-dimensional space" (p. 1425). Our guidance here is that researchers should choose the family of algorithms that makes the phenomenon legible and should demonstrate this with concrete examples to foster trustworthiness.

Second, parameterization here means theorizing by rendering. Hence, the final model may be chosen based on the meaningfulness of categories, dimensions, and their relationships for capturing the phenomenon of interest. Researchers begin with sensible defaults, but in this case, probe nearby specifications to see which rendition most clearly exposes the phenomenon. For example, Giorgi et al. (2019) started from a default value, as recommended by the tool, and proceeded to rely on metrics to find an optimal range of topics. Their final choice was, however, based on manual inspection of the surfaced topics. This shows how computational metrics might be helpful to determine an adequate range of values for parameters, but the final choice is informed by the researchers' interpretation. This is important because models that score well on computational metrics may not always be interpretable, and vice versa (Chang et al., 2009). This is also evident in how Miranda et al. (2022b) explained their choice of parameters through a sequence of steps. For instance, in explaining their choice among alternative models, they explained why they selected "a 23-topics model spanning four thematic quadrants" (p. 1425), elaborating why they did not just follow "fit metrics such as log-likelihood and perplexity" (p. 1446). Additionally, they utilized multidimensional scaling that enabled abductive sensemaking to foster trustworthiness.

Again, the focus on content versus structure matters in modeling choices. When content is central, researchers should choose models and settings that expose fine-grained meanings (e.g., a higher number of categories when warranted, tighter word-in-topic priors), preserve informative phrases and local context, and apply conservative normalization so that verbs, objects, and named entities remain legible. When structural attributes carry the theoretical claim, researchers should favor representations that explicate association structure and comparability across sources and demonstrate attribute revelatory potential. Researchers should pick the specification that allows for the most revelatory ordering, dispersion, and trajectory patterns among alternatives.

Accordingly, studies giving primacy to practice lexicons report alternatives to illuminate revelatory variance rather than merely to confirm stability. For example, Croidieu and Kim (2018) were aware of the sensitivity of the algorithm and tested topic stability across multiple runs. While they aimed to demonstrate stability, our framework invites reflection on the revelatory value of excluded models and topics as well. In this spirit, DCS researchers can draw inspiration from Sidorova et al. (2008), who utilized LSA to discover the intellectual core of the IS discipline. Although their goal was not necessarily to construct a theory, their

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analytical moves are inspirational in how they surfaced revelatory patterns. While they applied LSA on abstracts from articles published in IS journals, they tested models with different numbers of factors, not to show stability per se, but to enable more detailed analyses and theorizing. This illustrates reflexivity when it comes to interpreting different varieties of models while still acknowledging that varieties exist and reporting on them.

Because interpretations and decisions during model building are driven by the researchers' understanding of the phenomenon, this often also requires extensive domain expertise. Researchers giving primacy to practice lexicons are therefore more likely to complement the use of DCS tools with fieldwork or collaborate with domain experts. For example, Giorgi et al. (2019) relied on a strategy "that featured a triangulation of qualitative and quantitative analyses (a historical case study coupled with computational quantitative text analysis) to develop grounded theorizing" (p. 808). To facilitate trustworthiness, researchers can report on not only how different approaches complement each other (as in a mixed methods study) but also how the research design shapes the use of DCS tools.

Validating Novel Theoretical Patterns

After rendering an initial model under the primacy of practice lexicons, validation moves ask whether the new categories, relations, and claims credibly capture the phenomenon and offer novel theoretical insights. First, in establishing internal validity, trustworthiness starts by making the path from data to construct inspectable—i.e., face validity in the language of the practice, with provisional, practice-anchored labels. Researchers can show: (1) compact examples anchoring each category (top words and top documents), (2) how ambiguous cases were handled, (3) where splitting or merging of categories was enacted, and (4) why categories were dropped. For instance, Croidieu and Kim (2018) explained their coding procedure and visualized the mapping from topics to first-order concepts, second-order themes, and aggregate dimensions. Giorgi et al. (2019) dropped topics based on their understanding of the practice lexicon, strengthened internal validity through a systematic coding procedure (two independent coders), and provided genre-aware anchors (quotes from hearing exchanges and corporate passages). Miranda et al. (2022b) provided visuals for both synchronic and diachronic analyses. In all these cases, interpretive analytical moves such as independent memoing and reconciliation among authors are useful to demonstrate trustworthiness.

The goal of such illustrations is not only to provide proof that a certain category, dimension, or distribution can be considered explanatory, but also to invite readers to the qualitative interpretations of those patterns and their importance. To this end, visualizations are often supported by examples from the data in the form of raw quotes. When the focus is on the content, these quotes directly demonstrate the theorized categories. For instance, to validate their claims Croidieu and Kim (2018) provided several quotes from their raw data, thereby enabling readers to assess their interpretations of emerging categories. Alternatively, when the focus is on the structure of the text, direct quotes can demonstrate structural similarities or differences. Miranda et al. (2022b) explicitly compared two illustrative data excerpts from two actors: a journalist for the *Financial Times* and an IBM blockchain business development leader and social influencer, to compare the frames these actors used in their discourse on blockchain. Such reflection is especially important for studies giving primacy to practice lexicons because these diverse interpretations are potentially theoretically novel. Researchers need to convince readers that emerging categories or dimensions are indicative of a new theoretical concept.

Second, while the main focus for researchers giving primacy to practice lexicons may be on internal validity, there are still ways in which they can also illustrate external validity. One route is category sequencing with event anchoring—linking shifts in categories or attributes to recognizable field episodes (e.g., regulatory actions, product launches, scandals). A second route is to compare emerging patterns with existing measures, patterns, or trends associated with the phenomenon. For example, Croidieu and Kim (2018) compared their own results with two established measurement approaches to show that amateur radio operators became legitimized over time, similar to what they observed based on their own analysis. While their main contribution lies in providing "new explanations to understand the growing acceptance of non-professional expertise" (p. 2) in the form of what they call "lay-expertise legitimation," this analysis did help them validate their interpretations of legitimacy and, hence, the phenomenon under focus.

Lexical Alignment Through Iterative Sensemaking

As with studies with primacy to theoretical lexicons, studies giving primacy to practice lexicons align practice and theoretical lexicons iteratively across data work, model building and pattern surfacing, and validation and theorizing. The difference is emphasis: Researchers hold onto practice lexicons as the anchor and progressively construct theoretical lexicons that align with them. After an initial cycle, researchers can find that surfaced patterns largely reproduce established dimensions yet hint at novel ones. In such cases, they can return to data work (e.g., purposefully sampling cases that speak to the new pattern) or revisit model building (e.g., probe a nearby number of categories/priors or switch algorithms) to give voice to practice-anchored meanings.

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These returns are not distractions; they are how abductive sensemaking renders the phenomenon analytically tractable. Our framework asks authors to report a concise iteration trace: (1) what practice-anchored cue motivated a return, (2) the specific move enacted, (3) the observed effect on content or structure, and (4) why the post-iteration state better captures the phenomenon and how the theoretical lexicons were updated. Making these cycles visible shows not only where the analysis landed but how practice and theory were brought into correspondence step by step.

Discussion

In this methods article, we developed a framework for fostering trustworthiness in DCS research, where scholars construct theories by leveraging computational tools for dimensionalization and category surfacing. As shown in Figure 1, the framework turns on two key research design choices that govern both conduct and disclosure of corresponding analytical moves: (1) whether a study gives primacy to theoretical or to practice lexicons, and (2) whether the content of texts or the structure of the corpus is central to theorizing. Conduct concerns how researchers design and enact data work, model building and pattern surfacing, validation and theorizing, and iterative sensemaking; disclosure concerns the proportionate reporting of those moves. By placing trustworthiness at the center, the framework provides a structured way for authors, reviewers, and editors to assess the fit between objectives, analytical moves, and evidence in DCS research. Trustworthiness rests on researchers' reasoning about why particular moves are fit for purpose and in proportionate disclosure of that reasoning; it invites methodological bricolage oriented to the study's aims (Harley & Cornelissen, 2022; Lincoln & Guba, 1985; Pratt et al., 2020, 2022) and aligns with active categorization as a core engine of theory building (Grodal et al., 2021).

Our framework helps DCS researchers navigate the trustworthiness tensions that we identified earlier (see Table 1). We clarify that rather than inherently favoring one side over the other, each tension should be navigated depending on the study's research design. What constitutes trustworthy conduct and disclosure in each phase depends on whether a study gives primacy to theoretical versus practice lexicons, and on whether the content of texts versus the structure of the corpus is central to theorizing.

In data work, researchers strive to foster trustworthiness through rigorous practices but face a key dilemma as to whether to prioritize objectivity or customization in crafting these analytical moves. Our framework suggests that neither route is fundamentally better, and objectivity versus customization depends on research design choices. Hence, theory-primacy studies are expected to condition data work to the theoretical lens and theoretical standards, whereas practice-primacy studies are expected to preserve vernacular usage and corpus-specific expressions. In parallel, our framework distinguishes when content-focused studies should favor customization that preserves meaning, and when structure-focused studies should use uniform preprocessing and stable windows for structural comparability.

Similarly, in model building, researchers are often torn between seeking stability or variance. DCS tools can be tuned to become more stable (e.g., Yang & Subramanyam, 2023) but can also be leveraged to explore diverse categorizations to reflect upon. Our framework helps navigate this tension by arguing that theory-primacy studies should prioritize evidence correspondence to the guiding theoretical lexicon and pursue theoretical robustness, whereas practice-primacy studies should prioritize fidelity to practice lexicons while exploring revelatory models. Simultaneously, it highlights that choices should favor semantic distinctiveness when content is in focus, versus comparability and associations when structure is in focus. In all cases, researchers are encouraged to reflect on how patterns persist or shift under nearby configurations.

Finally, for validation and theorizing, our framework offers guidance for authors who struggle to validate models by either benchmarking against ground truth or accounting for interpretations. We recommend that theory-primacy studies validate in correspondence with extant constructs and theoretical frameworks, while practice-primacy studies validate against corpus-grounded meanings and practice accounts. In parallel, our framework clarifies when interpretive alignment should carry the evidentiary burden (focus on content) versus when structural patterns should be the focus of validation (focus on structure).

Overall, our framework encourages aligning conduct and disclosure to the study's objective, as shown in Figure 1. Further, Table 3 consolidates these recommendations into guidance for authors, editors, and reviewers. In the following, we first discuss the specific implications for DCS researchers before concluding with the implications for the CTC community more broadly.

Implications for DCS Research

Implications for Researchers

Our study offers several implications for researchers leveraging DCS tools to build theory by helping them navigate

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trustworthiness. First, the framework legitimizes distinct yet equally valid research designs when leveraging DCS tools to construe theories, and clarifies how to justify practices within each. Prior research catalogues recommended practices (e.g., Aceves & Evans, 2023; Hannigan et al., 2019; Kobayashi et al., 2018; Schmiedel et al., 2019), but researchers may lack guidance on how to choose among those practices. By grouping those practices under higher-order analytical moves and linking these moves to a study's pathway (primacy of theoretical vs. practice lexicons) and focus (on content of texts or structure of the corpus), the framework can help researchers make informed choices. For each type of design, the framework provides specific guidelines for conduct and disclosure.

Second, the framework establishes links between achieving rigor and fostering trustworthiness in DCS research. While existing journal guidelines on transparency have strengthened rigor in scholarship (e.g., Burton-Jones et al., 2021; DeCelles et al., 2021), they take for granted the distinctive trustworthiness tensions faced by DCS researchers. Thus, instead of thinking of our framework as a template, we encourage researchers to leverage it as a "canvas" that explains how different pieces of a puzzle come together (Dorobantu et al., 2024). Trustworthiness is achieved by demonstrating that the enacted moves are commensurate with the chosen research design and by providing just enough evidence to make the reasoning inspectable. In other words, our framework is an invitation to researchers to construct their own methods such that the community recognizes such methodological bricolage as legitimate.

Third, our framework foregrounds the inherently iterative character of DCS research. To ensure ease of understanding, we have presented our framework in a format that researchers can relate to as they move through research phases of data work, model building and pattern surfacing, and validation and theorizing. Yet we have argued and demonstrated that the process of theory construction using DCS tools is an inherently iterative endeavor. Theory construction does not happen merely at the last stage of research; it is simply a milestone that marks completion of an iteration and triggers the next iteration in the process of knowledge production. Iterations are fundamental for researchers dealing with the challenges of the lack of ground truth in and the opacity of DCS research until they achieve their goals of theory construction (see Miranda et al., 2022a for a discussion on stopping rules). Our framework offers a glimpse of what such iterations within and across the phases will look like and encourages researchers to be reflexive about them. DCS researchers should embrace these iterations, consider how they may have shaped their theory construction processes, and report on those iterations that were consequential to the theory they ended up constructing. This will help to foster trustworthiness not only during the review process but also for the readers who draw on these studies for the purpose of knowledge building.

Table 3. Guidance for Fostering Trustworthiness in DCS Research

Giving primacy to theoretical lexicons	Giving primacy to practice lexicons
Theory-driven data work *Conduct (overall)* - Use theory-anchored research questions to script sampling and perform preprocessing as pre-analysis active categorization. *Conduct (content versus structure focus)* - Foreground texts that narrate concrete meanings tied to focal constructs while preserving semantic distinctiveness in data preprocessing (when content in focus); prioritize structural signal and comparability via balanced sampling and theory-implicated data preprocessing (when structure in focus). *Disclosure* - Specify and justify all preparation steps in relation to theoretical aims, explicitly reflecting on what constitutes data, and distinguish theory-motivated steps from tool defaults.	**Phenomenon-driven data work** *Conduct (overall)* - Use research questions grounded in focal phenomenon to script sampling and perform preprocessing based on understanding of practice lexicon. *Conduct (content versus structure focus)* - Foreground texts that narrate concrete meaning-bearing practices and discourses, while preserving semantic distinctiveness in preprocessing (when content in focus); protect structural signal and comparability while relating rich metadata to data preprocessing (when structure in focus). *Disclosure* - Specify and justify all preparation steps in relation to how they clarify the phenomenon under study, explicitly reflecting on what constitutes data, and make explicit how phenomenon-specific needs depart from generic preprocessing standards.

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Surfacing theoretically robust patterns	**Surfacing empirically revelatory patterns**
Conduct (overall) - Select algorithms from prior studies with similar contexts, theories, and data, being mindful of underlying assumptions inscribed in those algorithms and selecting configurations that are theoretically meaningful. *Conduct (content versus structure focus)* - Favor precise semantic splitting to preserve distinct theoretical meanings (when content in focus); favor relating categories to covariates to maintain comparability across strata and alignment with theory (when structure in focus). *Disclosure* - Explain why the chosen algorithm family fits the intended purpose and how theoretical assumptions are aligned; report and justify parameter choices beyond the obvious; demonstrate how patterns persist or shift under nearby configurations.	*Conduct (overall)* - Choose algorithms that make the phenomenon legible, often leveraging domain expertise, being mindful of underlying assumptions inscribed in those algorithms and selecting the rendition that most clearly reveals the phenomenon. *Conduct (content versus structure focus)* - Favor models and settings that expose fine-grained, practice-rooted meanings (when content in focus); use representations that explicate association structure yielding the most revelatory patterns among alternatives (when structure in focus). *Disclosure* - Show why the chosen algorithmic approach illuminates the focal dynamic; report informative parameter specifications to illuminate revelatory variance; reflect on the revelatory value of included and excluded models and categories.
Validating patterns against existing theory	**Validating novel theoretical patterns**
- *Conduct (overall)* Establish internal validity by showing that the category system instantiates theoretical constructs through face validity; strengthen external validity via triangulation with independent data/methods, category-sequencing with event anchoring, and construct alignment with preexisting frameworks. *Conduct (content versus structure focus)* - Show label/interpretation stability under reasonable alternatives and anchor in text, then compare to established constructs/taxonomies (when content in focus); demonstrate attribute invariance and document that alternatives do not alter the structural signal (when structure in focus). *Disclosure* - Visualize top words and documents with provisional, theory-anchored labels; report intersubjective interpretability; add quotes to justify theory-informed interpretation; show structural metrics invariance under reasonable perturbations.	*Conduct (overall)* - Establish internal validity by making the path from data to construct inspectable through face validity; strengthen external validity via category-sequencing with event anchoring or comparisons of emerging patterns to existing phenomenon-linked measures. *Conduct (content versus structure focus)* - Use raw quotes that directly demonstrate the theorized categories and invite qualitative interpretation of their relevance for novel concepts (when content in focus); use direct evidence and link shifts in categories/attributes to recognizable field episodes to validate structural patterns (when structure in focus). *Disclosure* - Visualize the mapping from categories to aggregate theoretical dimensions; provide examples and raw quotes anchoring each category; show how ambiguous cases were handled and where categories were split, merged, or dropped; support visuals that demonstrate theorized categories or structural patterns; and use memoing and author reconciliation to demonstrate interpretive trustworthiness.

Implications for Reviewers

The framework also serves as a sensitizing device for reviewers tasked with evaluating papers using DCS tools for theory construction. We recognize that on many occasions, reviewers are invited to serve on a DCS manuscript because of their proficiency with computational tools, either through their own use or contributions to tool development. Given their expertise with specific tools, such reviewers bring valuable methodological insights. Yet evaluation in DCS research does not merely depend on expertise in specific tools; rather, it should focus on whether the analytical moves enacted by authors are fit for the study's stated objective.

The framework is especially useful for reviewers who know the tools but do not use them for constructing theory, or who have approached theory construction with DCS tools from only specific research designs in their own research (as summarized in Table 2). In practice, this means reading for a clear sense of the research objective of a given study and then

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asking whether the evidence matches the burden implied by that objective. Reviewers should, for example, ascertain: When content is in focus, do the authors establish face-valid labels anchored in text and show how meanings persist or vary under reasonable nearby choices? When structure carries the load, do they keep the basis of comparison the same across groups or time (same corpus boundaries, units, preprocessing, windows) and anchor patterns to recognizable events where appropriate? Reviewers should avoid pushing authors toward conflicting analytical moves—for example, holding a study that prioritizes practice lexicons (aimed at discovering novel theoretical patterns) to the same robustness rituals as a study that prioritizes theoretical lexicons (aimed at explaining a phenomenon through a chosen theory) is unfitting and unlikely to produce a developmental review.

The framework thus invites reviewers to recognize different approaches—giving primacy to theoretical lexicons versus practice lexicons and focusing on content versus structure—as legitimate routes to interpretive, iterative theory construction with DCS tools. Methodological pluralism remains the standard, but rigor is calibrated: assess whether the enacted moves and the evidence shown are commensurate with the stated objectives, not whether the paper mirrors rituals from other genres. A clear iteration trace should make consequential iterations visible (why the authors went back, which move changed, what shifted in content or structure, and why the new state better instantiates the theory or captures the phenomenon).

Finally, reviewers should evaluate DCS research on its own terms—a specific method family within the CTC research genre that is neither purely quantitative nor qualitative, neither purely theory- nor phenomenon-driven, and neither purely content- nor structure-focused (Berente et al., 2019; Miranda et al., 2022a). We invite reviewers to use the framework as a canvas for developmental feedback that advances trustworthiness rather than defaulting to checklists borrowed from other traditions.

Implications for Editors

Our framework also has implications for the senior and associate editors handling DCS research-based manuscripts; these editors not only shape discourses at the level of the manuscript but also at the field level, in the spirit of cumulative knowledge building in DCS research. At the microlevel, our framework may help in several ways. For instance, editors can use our framework to orchestrate the matching process between researchers and reviewers. Ideally, editors would assign submitted studies to reviewers who are likely to follow a similar approach to constructing theory in their own research. But, even when such an ideal matching is not possible (which is more likely given the nascent nature of the community), editors can use our framework to educate reviewers on their expectations, both up front and during the ongoing exchanges between reviewers and researchers. We encourage editors to intervene whenever they see reviewers pushing researchers in the wrong direction or researchers failing to leverage the power of computational tools in constructing theory. In other words, by using our framework, editors can make sure that authors follow appropriate analytical moves in line with their chosen objectives and that reviewers do not fall into the trap of the binary distinctions. The key objective would be to foster trustworthiness in the review process.

At the field level, our framework speaks to editors' broader responsibility to manage the transparency agenda within their journals, considering the potential costs and dysfunctional side effects that rigid policies might impose, such as marginalizing certain research communities or methods. By adopting a framework tailored to the specific family of DCS research, editors can advance trustworthiness without inadvertently hindering the development of this important research area. They can encourage in-depth discussion sections, where researchers reflect honestly on what was learned. As the field evolves, editors can adapt their policies based on learnings from researchers, reviewers, and the experiences of other journals. Our framework provides a foundational step in characterizing what fostering trustworthiness means in DCS research. Our hope is that future editorials about rigor, trustworthiness, or transparency in leading journals will conceive CTC as a unique research genre (and DCS research as a unique method in that genre) that requires specific guidelines of its own.

Implications for Maneuvering Hybrid Designs and Switching Moves

Our framework assumes that DCS studies fit into unique combinations of the two axes—primacy of theoretical versus practice lexicons, and focus on the content of texts versus the structure of the corpus. We made this distinction for analytical purposes. In reality, it is possible that DCS studies could be more hybrid. First, researchers may pursue both category meanings rooted in content and corpus-level structure—using surfaced categories to interpret meaning while also leveraging distributions, associations, or trajectories to make comparative or temporal claims. Second, as analysis unfolds, researchers may iterate not only across research phases but also across the framework's two axes—shifting between the primacy of theoretical lexicons and the primacy of practice lexicons, and, at times, shifting between a focus on content and a focus on structure. Our framework treats hybridity and switching as

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legitimate features of DCS research, but it also clarifies their implications for trustworthiness: They change what ought to be demonstrated and disclosed. A basic discipline is therefore required. Authors should be explicit about what carries the theoretical load in the focal claim—whether the claim relies primarily on category meanings or on corpus-level structure—and avoid validating one kind of claim with the evidentiary logic of the other. When a study advances both kinds of claims, authors should either specify which claim is primary and treat the other as corroborative evidence, or treat them as distinct claims with distinct validation targets; in either case, conduct and disclosure should be aligned to the evidentiary logic of each rather than blended in ways that make the contribution difficult to evaluate.

Relatedly, researchers should be aware that iterations can take shape between the pathways of primacy to theoretical and practice lexicons, and can also involve shifts in whether the study is content-focused or structure-focused. For example, researchers may start by granting primacy to theoretical lexicons by engaging in theory-driven data work and building models intended to surface theoretically meaningful categories. In the process, however, patterns may emerge that do not align with the initial theoretical lexicons. At this stage, researchers may decide to bracket those patterns and continue pursuing the original objective, or they may decide to switch and give primacy to practice lexicons to explore the phenomenon in depth. Similarly, researchers may begin with a content focus (treating category meanings as load bearing) and later realize that the theoretical contribution rests more plausibly on corpus-level structure (or vice versa), prompting changes to sampling, preprocessing, parameterization, and validation. In such scenarios, switching often entails *redoing earlier phases*—especially data work and model building—and, crucially, respecifying the burden of proof by stating what now carries the theoretical load and what validation targets follow from that choice. We note that such switching is not hypothetical; our interviewees described multiple narratives of switching in DCS research. Yet in final published versions of manuscripts, these shifts are often underreported. Based on our framework, we encourage authors to disclose how, when, and why they switched and to walk readers through the narrative of respecifying aims and realigning analytical moves. One practical way to do so is to include a brief switch trace that states the trigger for the switch, what axis shifted (lexicon primacy and/or content/structure focus), what analytical moves were revised in respective phases, what changed in the surfaced output, and why the post-switch configuration better supports the study's focal claim. Accordingly, where a manuscript reports a mid-project switch or hybrid approach, reviewers should look for a clear account of the trigger, the rescoped data-work rules, and how the alignment between aims, moves, and evidence was restored.

More research is required on hybridity and switching in DCS studies—particularly on how often such switches occur, what triggers them, and how they reshape the burden of proof across data work, model building, and validating. Future work could also examine whether some switching patterns become routinized within research communities, and how to strengthen trustworthiness in these studies without having to write prohibitively lengthy manuscripts.

Implications for the Broader CTC and Computational Research Communities

Our framework speaks most directly to researchers who use DCS tools with an interpretive orientation to construct theory from text. In developing the framework, we have already explained how interpretive DCS research may draw inspiration from adjacent research that leveraged DCS tools for other purposes. At the same time, several of the core ideas of our framework travel beyond the specific domain of DCS research and open avenues for future work on the use of computational tools in research more generally. The underlying principle is to make research objectives, analytical moves, and proportionate disclosure commensurate—whatever the specific tool—so that claims are rendered trustworthy for the purpose at hand. Here, we discuss such implications and opportunities for future research within the broader CTC and computational research communities.

A first implication concerns the use of large language models (LLMs), a specific class of generative AI models that can process and generate text, in constructing theories. LLMs are similar to the DCS tools we have discussed in that they operate on text and are powered by ML algorithms, but they are importantly different: They arrive as pretrained, ready-to-hand, generative models, rather than as algorithms that researchers use from scratch to build context-specific models. Hence, the question is what happens when some choices that researchers make in constructing their models using DCS algorithms are now prebuilt, in the case of LLMs?

There are several considerations for trustworthiness if researchers use prebuilt LLMs instead of the typical DCS tools. For example, the use of LLMs would shift the weight of choices and thereby trustworthiness to different activities. With such prebuilt models, researchers need to grapple with an ensemble of algorithms where some of the key choices that researchers used to make in data work and model building—e.g., data preprocessing, selection of algorithms, hyperparameter tuning—are now pre-crafted in a ready-to-use model. These choices still happen, but in the background, often without the control or knowledge of the researcher. Hence, the weight of trustworthiness shifts from preprocessing data and building models to tweaking model

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inputs and outputs and crafting prompts. This has parallels with the concept of envelopment, where the researchers may control the context, inputs, and outputs, but the inner workings of the model may remain black-boxed (see Floridi, 2011). Additionally, validating surfaced patterns becomes more important than ever, due to the increased opacity and lack of control over how the model has been trained.

Further, the use of LLMs is likely to expand the scope of expectations for research activities that require trustworthiness. Our framework focuses on fostering trustworthiness in CTC across the phases of data work, model building and pattern surfacing, and validation and theorizing. If researchers start using LLMs, we may need to expand the trustworthiness expectations to other research activities, including how and whether LLMs were used for idea generation, summarizing the literature, and writing the manuscript, among others. The question then is not only about whether the knowledge claims in a particular study are reliable and valid but also whether the knowledge claims are original. It would be a very interesting and important avenue for future research as we enter the new age of computational tools. The CTC community will benefit from cumulative attempts at proposing new ways of thinking about these issues. In this spirit, our framework is a starting point, perhaps a first step, in helping characterize what fostering trustworthiness means in CTC research.

It is important to note that we recognize the growing debate around whether researchers should use LLMs and other generative AI applications for their research (see Bechky & Davis, 2025; Davison, 2025; Gopal et al., 2025; Kulkarni et al., 2024; Susarla et al., 2023). This debate is beyond the scope of our study. We do not take a position on whether researchers should use these tools; rather, we clarify how the core logic of our framework may still help researchers in fostering trustworthiness if they do.

The second implication concerns computational tools that go beyond text-centric DCS tools. Theories can be constructed by leveraging computational tools that may not require the analysis of textual data—for example, analytical modeling (Sahaym et al., 2023), relational event modeling (Lindberg et al., 2024), and computational simulations (Reider et al., 2024). Such tools and associated research designs are much different from the use of DCS tools that our framework explains. We argue that in such cases of CTC research, the community will still benefit from embracing trustworthiness instead of transparency, although what aspects of research require trustworthiness and how to foster trustworthiness in those aspects will need more work. Our framework will first need to be extended before it is applied to these contexts. Further research could build on our key argument that there are different objectives for using the same family of tools and that analytical moves differ depending on those objectives. Researchers could then engage in a more in-depth empirical analysis of what those analytical moves are exactly (as we did through our methodological review and interviews). For instance, one could use computational simulations to formalize theories in two ways: either by formalizing theoretical insights generated inductively from observed data or by formalizing a deductively developed verbal theory and creating synthetic data. Both objectives are feasible, both will entail construction relationships through computational simulations, but both will require a different set of analytical moves and hence distinct expectations for fostering trustworthiness.

Our final implication is for computational studies that primarily test theories using DCS tools. Given that our framework only speaks to the use of DCS tools for theory construction, we left out theory testing objectives. Yet we argue that these studies may still benefit from our framework. Consider the study by Kaplan and Vakili (2015), in which the authors used topic modeling to operationalize a construct of breakthrough innovation. Their primary aim in using the DCS tool was that it afforded a measurement of a construct in the service of theory testing. We contend that such studies are close to the pathway of giving primacy to theoretical lexicons, as per our framework. Even though their goal is to test theories, it is likely that unexpected patterns will surface due to the unsupervised nature of the tools they use. Such patterns may not fit with their predefined theoretical constructs but may help them further develop their theory. Hence, they may switch from the original idea of theory testing to incorporate some elements of theory construction with DCS tools. If they do so, they may be able to leverage our framework in demonstrating trustworthiness. A good example of this possibility is Bao and Datta (2014). They employed topic modeling for quantifying risk types from textual risk disclosures to test their theory, but in doing so, they were also open to discovering novel risk types. Their paper started with the primacy of theoretical lexicons. But when they witnessed novel patterns surfacing from the practice lexicons, instead of ignoring those patterns, they embraced them. They were able to fit these two approaches into one manuscript and offer a methodological contribution.

We call for researchers to embrace this possibility of simultaneous theory testing and construction in their empirical papers, and our framework may offer some insights on how to do that. Our guidance here is simple: If a switch occurs, document the trigger, the rescoped analytical moves for data work and modeling, and how the alignment among aims, moves, and evidence was restored. Doing so lets a single manuscript host both testing and construction without conflating their evidentiary burdens, and it renders the resulting contribution trustworthy on its own terms.

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Conclusion

CTC is a new and emergent research genre, and DCS research is a prominent method family within CTC. What we do in the community today is going to have a lasting influence on how we, as CTC researchers, contribute to IS, management, and organizational scholarship. The initial manuscripts that are being published now are going to shape the field for decades. At this crucial juncture, it is important to take a step back and reflect on how we conduct our research practices and how we disclose our choices in published manuscripts. Our study is an attempt in this direction, wherein we propose trustworthiness—rather than inherited transparency checklists—as the evaluative anchor for theory construction with DCS tools. Our framework advances shared evaluative logic for researchers, reviewers, and editors that preserves methodological pluralism while enabling cumulative theorizing. Rather than aspiring to a single correct rendering, the field can converge on a simple commitment: declare what kind of claim is being made, enact moves that are fit for that claim, and provide proportionate evidence that the claim holds under reasonable alternatives. We are hopeful that our framework will meaningfully inform further CTC research in IS, management, and organizational scholarship.

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